26. Communication without internet: AM, FM, long wave and two-way radio (English)

Oracle OS guide. See also the Oracle **Radio** screen (official stations per country).

Listening — no licence needed

- **FM (87.5–108 MHz)**: tens of km. Phones with FM need **wired headphones** (antenna); works without SIM, internet or in airplane mode.
- **AM / medium wave**: hundreds of km, especially at night; needs an AM radio (most phones lack it).
- **Shortwave**: international news.
- **Long wave**: BBC Radio 4 on 198 kHz was switched off on 27 June 2026.
- A **wind-up/solar AM/FM/SW radio** is the most reliable kit item. Save batteries: listen **on the hour** for news.
- In a UK national emergency the BBC (Radio 4 and nations/local stations such as BBC Radio Scotland) carries government announcements.

Talking — two-way radio

- **PMR446 (Europe/UK)**: licence-free walkie-talkies, 446 MHz, 0.5 W, 16 channels, typically 1–3 km in towns. Agree a **channel and schedule** with family (e.g. channel 3, 5 minutes every hour on the hour).
- **CB 27 MHz**: licence-free in UK/EU (CEPT); **channel 9** commonly used for emergencies, 19 for road traffic.
- **USA**: FRS (licence-free), GMRS (simple licence), CB channel 9.
- **Amateur radio**: **licence required**; amateurs run emergency nets (ARES, RAYNET) — see ARRL ARES manual in the library. In imminent danger to life, ITU Radio Regulations allow any means to call for help.
- **Marine VHF channel 16 (156.8 MHz)**: distress at sea.

Calling for help

"**MAYDAY ×3**" (danger to life) or "**PAN-PAN**" (urgent). Who you are, **where** (coordinates from Oracle's Emergency screen), what happened, how many people, what help. Speak slowly, repeat, **listen**.

Signals without radio

Three of anything = distress (3 whistle blasts, lights, fires). Alpine: 6 blasts per minute. Morse SOS ··· ––– ···. Ground-to-air (3 m letters): **V** need help, **X** need medical help, **N** no, **Y** yes, **→** going this way. Signal mirror or phone screen to reflect sunlight.

Mobile phone

112 works across Europe and in many countries on any available network; many phones call 112 without a SIM. SMS gets through when calls fail. Save battery: airplane mode between checks. Keep cell-broadcast emergency alerts enabled.